

Algorithms for Multi-Channel Aggregated Convergecast in Sensor Networks

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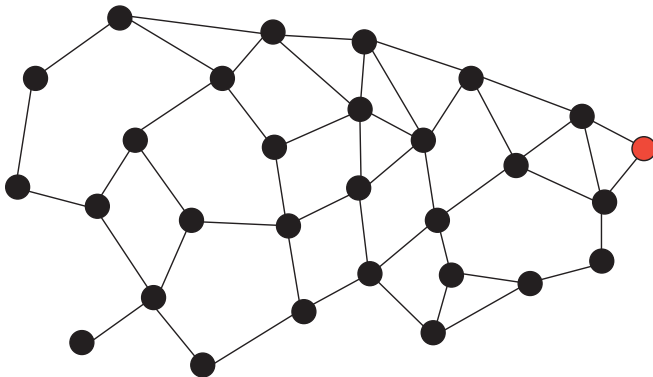
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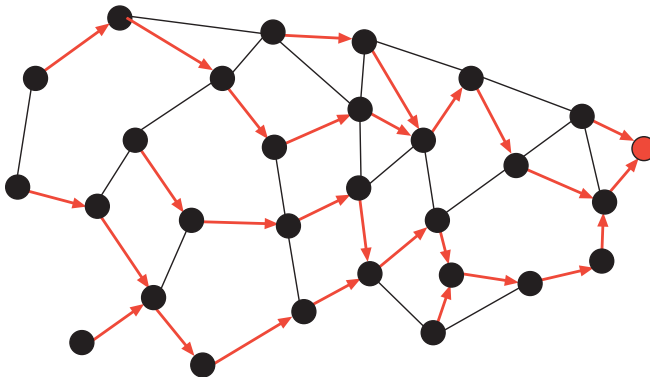
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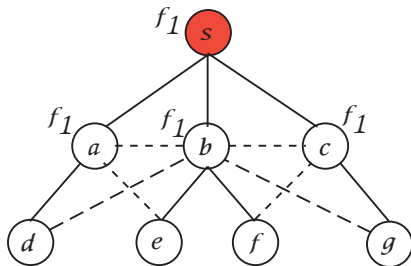
A Data Aggregation Network



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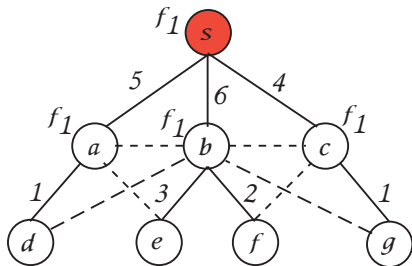
An Illustration



Setting

- (1) Half-duplex single transceiver
- (2) Nodes aggregate pkts from children and transmit only one pkt
- (3) TDMA
- (4) Interference causes pkt loss
- (5) Receiver-based frequency assignment

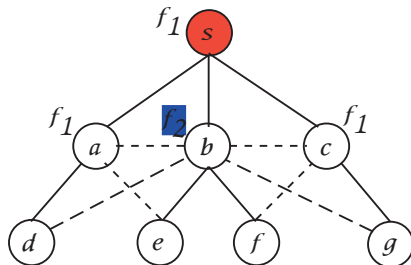
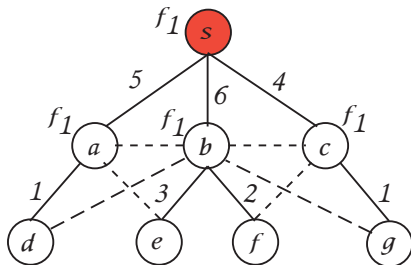
An Illustration



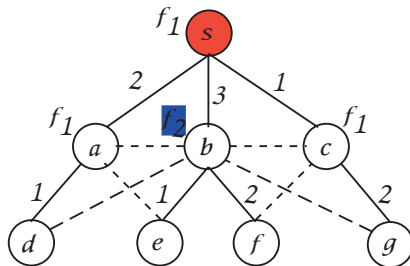
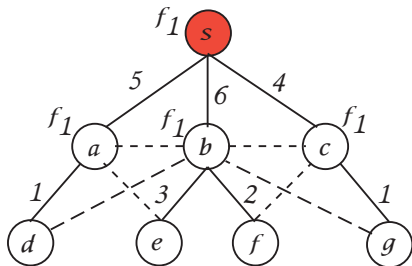
Setting

- (1) Half-duplex single transceiver
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An Illustration



An Illustration



Question

What is the fastest rate at which aggregated data can be collected from the network? (minimizing the schedule length)

	Frame 1			Frame 2		
Receiver	Slot 1	Slot 2	Slot 3	Slot 1	Slot 2	Slot 3
s	c	a,d	b,e,f	c,g	a,d	b,e,f
a	d			d		
b	e	f		e	f	
c		g			g	

Minimum Frequency Assignment Problem

MFAP

Given a spanning tree T on an arbitrary graph $G = (V, E)$, find the **minimum number of frequencies** that can be assigned to the receivers of T such that all the interfering link constraints are removed.

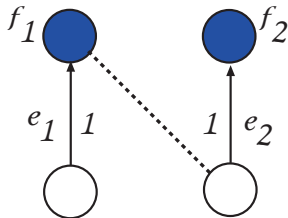


Figure: Interfering edge structure

Theorem

MFAP is NP-complete.

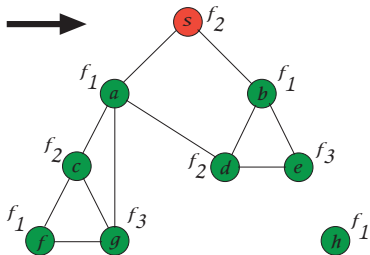
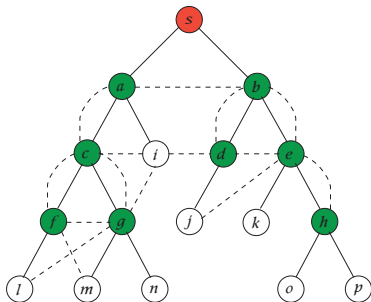
Proof: Reduction from Vertex Color.

An Upper Bound

Lemma

Construct a **constraint graph** $G_C = (V_C, E_C)$ from $G = (V, E)$ as follows:
For each **receiver** in G , create a vertex in G_C . Create an edge between two such vertices in G_C if their corresponding receivers in G are part of an interfering edge structure.

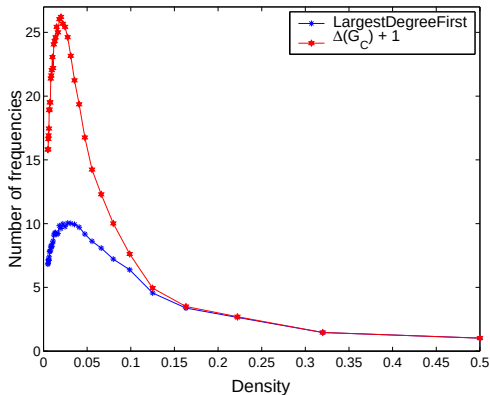
Then, the number of frequencies required to remove all the interfering link constraints is: $K_{\max} \leq \Delta(G_C) + 1$, where $\Delta(G_C)$: max degree in G_C .



Evaluation: Frequency Bounds

Largest Degree First (LDF)

1. while $V_C \neq \phi$ do
2. $u \leftarrow \text{max degree in } V_C$
3. Assign the first available frequency to u different from its neighbors
4. $V_C \leftarrow V_C \setminus \{u\}$



BFS Time Slot Assignment

Algorithm

Input: $T = (V, E_T)$
 while $E_T \neq \phi$
 $e \leftarrow$ next edge from T in BFS order
 Assign minimum time slot to e
 $E_T \leftarrow E_T \setminus \{e\}$

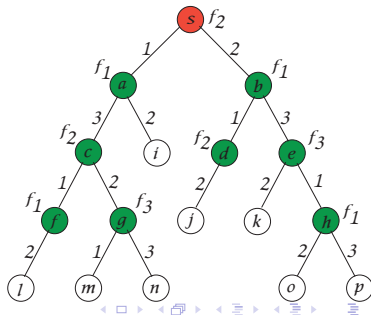
Theorem

Algorithm

BFS-TIME SLOT ASSIGNMENT on a tree gives the minimum schedule length equal to $\Delta(T)$, where $\Delta(T)$ is the maximum node degree in T .

Proof by induction on i :

$$N(i+1) = \max\{N(i), N(i) + 1\}$$



Multiple-Frequency Minimum Time Scheduling Problem

MFMTSP

Given a spanning tree T on an arbitrary graph $G = (V, E)$ and q frequencies, find an assignment of frequencies and time slots such that the schedule length is **minimized**.

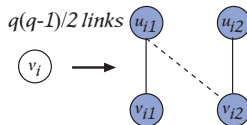
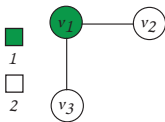
Theorem

MFMTSP is NP-complete.

Proof: Reduction from Vertex Color.

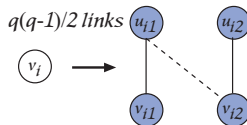
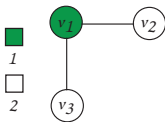
Multiple-Frequency Minimum Time Scheduling Problem

Reduction from Vertex Color.

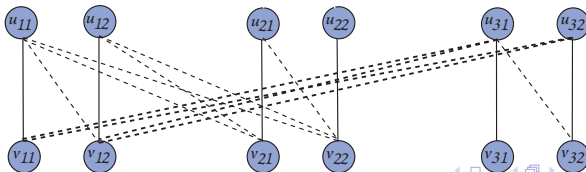


Multiple-Frequency Minimum Time Scheduling Problem

Reduction from Vertex Color.

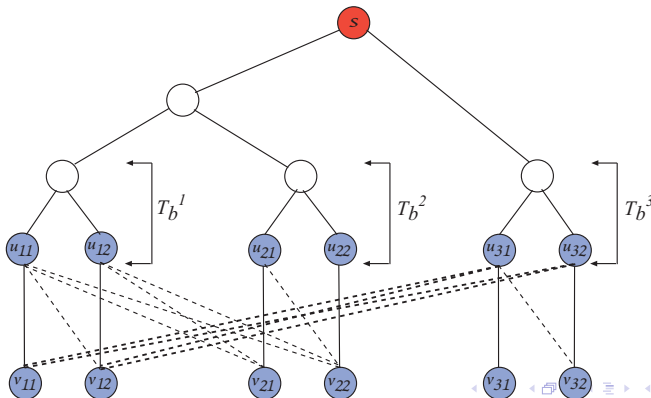
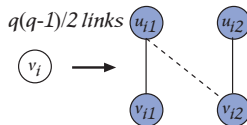
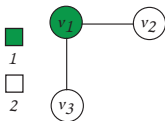


q^2 interfering links



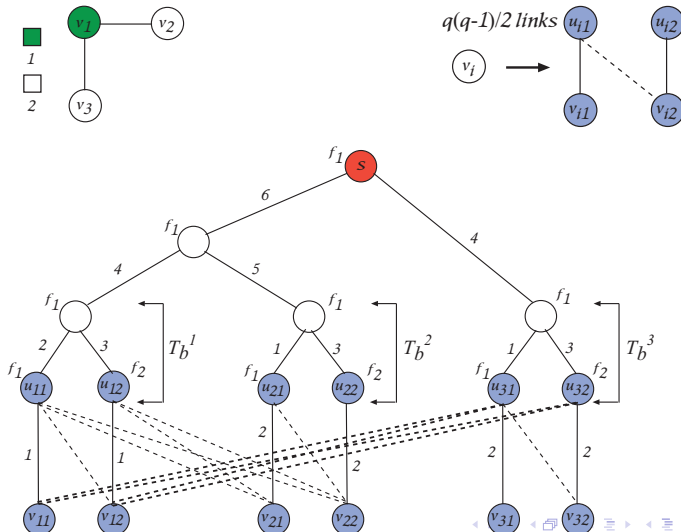
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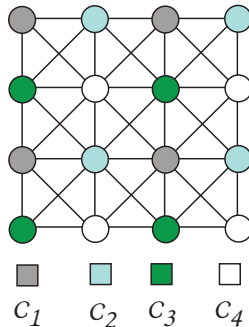
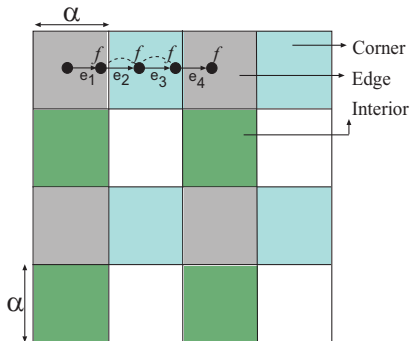


Time Slot Assignment on UDG

Lemma

Suppose γ_c denote the set of time slots to schedule the edges in cell c . Then, the minimum schedule length Γ for the whole network is:

$$\Gamma \leq 4 \cdot \max_c \{|\gamma_c|\}, \forall \alpha \geq 2.$$



Frequency Assignment on UDG

Load-Balanced Frequency Assignment

$R = \{v_1, \dots, v_n\}$: receivers on T

$m : R \rightarrow \{f_1, \dots, f_k\}$

If $m(v_j) = f_k$, then children of v_j transmit on f_k

Define **load** on f_k under m as:

$$I^m(f_k) = \sum_{m(v_j)=f_k} \deg^{in}(v_j)$$

Then, a **load-balanced frequency assignment** m^* is:

$$m^* = \arg \min_m \max_{f_k} \{I^m(f_k)\}$$

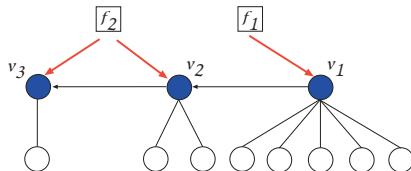


Figure: $I(f_1) = 5$, $I(f_2) = 5$

Frequency Assignment on UDG

Load-Balanced Frequency Assignment

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$m : R \rightarrow \{f_1, \dots, f_k\}$

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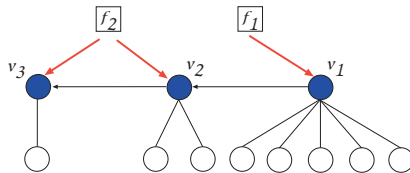


Figure: $I(f_1) = 5$, $I(f_2) = 5$

Lemma

Load-balanced frequency assignment is NP-complete.

OPT (min-max) load is L^{m^} .*

Frequency Assignment on UDG

Algorithm FREQUENCYGREEDY (ϕ)

(1) Sort the receivers in non-increasing order of in-degrees.

$$\text{deg}^{\text{in}}(v_1) \geq \text{deg}^{\text{in}}(v_2) \geq \dots \geq \text{deg}^{\text{in}}(v_n).$$

(2) Starting from v_1 , assign each successive node a frequency from $\{f_1, \dots, f_K\}$ that has the **least load**, breaking ties arbitrarily.

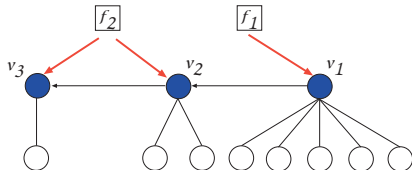


Figure: $l(f_1) = 5, l(f_2) = 5$

Frequency Assignment on UDG

Algorithm FREQUENCYGREEDY (ϕ)

(1) Sort the receivers in non-increasing order of in-degrees.

$$\deg^{in}(v_1) \geq \deg^{in}(v_2) \geq \dots \geq \deg^{in}(v_n).$$

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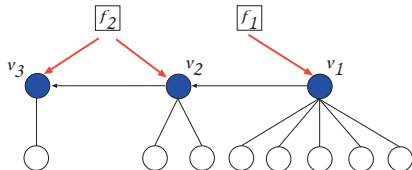


Figure: $I(f_1) = 5$, $I(f_2) = 5$

Lemma

Algorithm FREQUENCYGREEDY gives a $(4/3 - 1/3K)$ approximation on L^{m*} .

Proof: Follows from Graham's algorithm for scheduling jobs on identical parallel machines according to longest processing time first (LPT).

An Upper Bound on γ_c

Lemma

If L_c^ϕ denote the load on the *maximally loaded frequency* in cell c achieved by **FREQUENCYGREEDY**, then *any greedy* time slot assignment can schedule all the edges in c within $2L_c^\phi$ time slots, i.e.,

$$|\gamma_c| \leq 2L_c^\phi.$$

A Constant Factor Approximation

Theorem

Given a tree T on a UDG G , and K frequencies, there exists an algorithm $\mathcal{G}$ that achieves a constant factor $8\mu_\alpha (4/3 - 1/3K)$ approximation on the optimal schedule length, where $\mu_\alpha > 0$ is a constant for a given cell size $\alpha \geq 2$, i.e.,

$$\Gamma_{\mathcal{G}} = O(\Gamma_{OPT})$$

Proof Sketch

$\mathcal{G}$ consists of 2 phases:

- (1) Run **FREQUENCYGREEDY** in each cell c_i
- (2) Run any greedy time slot assignment scheme
 e.g., schedule a **maximal number of edges** in each slot

Lower bound on OPT: $\Gamma_{OPT} \geq \max_c \{L_c^{m*}\} / \mu_\alpha$

$$\begin{aligned}
 \Gamma_{\mathcal{G}} &\leq 4 \cdot \max_c \{|\gamma_c|\} \leq 8 \cdot \max_c \{L_c^\phi\} \\
 &\leq 8 \cdot \max_c \{(4/3 - 1/3K) \cdot L_c^{m*}\} \\
 &\leq 8\mu_\alpha (4/3 - 1/3K) \cdot \Gamma_{OPT}
 \end{aligned}$$

Arbitrary Trees Under UDG

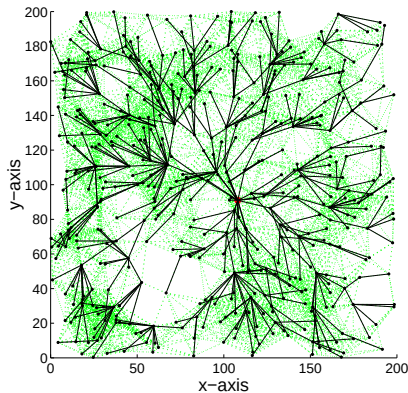


Figure: Shortest Path Tree

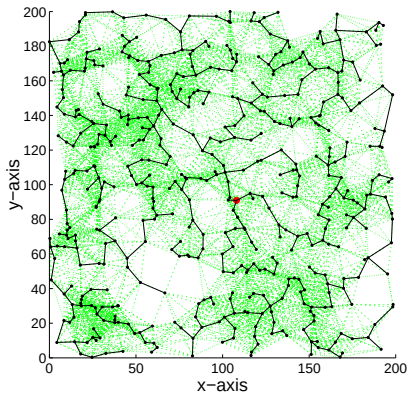


Figure: Minimum Spanning Tree

Arbitrary Trees Under UDG

Theorem

Given a UDG G and K frequencies, there exists an algorithm $\mathcal{H}$ that achieves a constant factor $8\mu_\alpha\Delta_C$ approximation on the optimal schedule length, where $\mu_\alpha > 0$, $\Delta_C > 0$ are constants for a given cell size $\alpha \geq 2$, i.e.,

$$\Gamma_{\mathcal{H}} = O(\Gamma_{\text{OPT}})$$

Proof Sketch

V_c : set of nodes in cell c

$R_c = \{v_1, \dots, v_n\}$: set of receivers in c on arbitrary T

$\Delta^{in}(T)$: max in-degree in T

Lower bound on OPT: $\Gamma_{OPT} \geq \max_c \left\{ \left\lceil \frac{|V_c|}{K} \right\rceil \right\} / \mu_\alpha$

Then

$$\begin{aligned} \Gamma_{\mathcal{H}} &\leq 4 \cdot \max_c \{|\gamma_c|\} \leq 8 \cdot \max_c \{L_c^\phi\} \\ &\leq 8 \cdot \max_c \left\{ \left\lceil |V_c|/K \right\rceil \right\} \cdot \Delta^{in}(T) \\ &\leq 8\mu_\alpha \Delta^{in}(T) \cdot \Gamma_{OPT} \end{aligned}$$

Bounded-degree spanning tree always exists on UDG. Therefore, proved.

Schedule Length

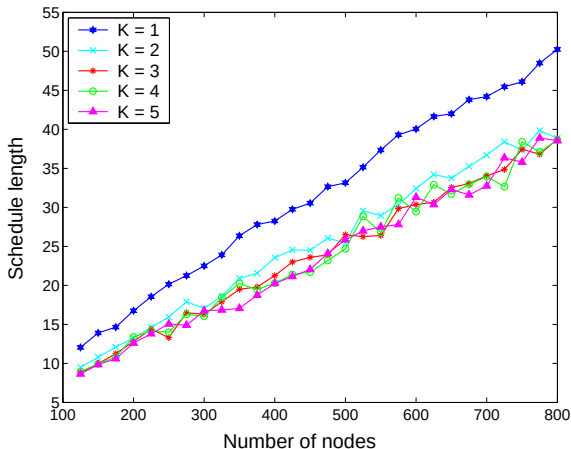


Figure: Algorithm $\mathcal{G}$ on Shortest Path Tree for Different Network Sizes

Schedule Length

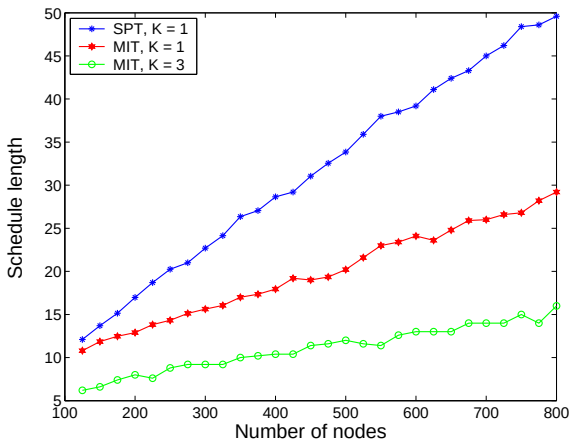


Figure: Algorithm $\mathcal{G}$ on Minimum Interference Tree for Different Network Sizes

Conclusions

Summary

- (1) Addressed a scheduling problem under aggregated convergecast using multiple channels.
- (2) Proved a NP-completeness result on finding the minimum number of channels required to remove all the interfering links in an arbitrary wireless network.
- (3) Proved a NP-completeness result on minimizing the schedule length for a given number of channels in an arbitrary wireless networks.
- (4) Proposed an optimal time slot scheduling scheme when enough frequencies are available.
- (5) Proposed constant factor approximation algorithms to minimize the schedule length on unit disk graphs.
- (6) Evaluated algorithms using simulations: most of the times 3 to 4 frequencies are enough all the interfering links.